and philanthropist Seth Framji Cowasji Banaji [5] to tide over the water crisis in Bombay. [6]

Four years later, on the eastern coast in Calcutta, Goluk Chandra built and demonstrated, without any Englishman's help, a stationary steam engine, and won a prize at the Agri Horticultural Society's meeting.

It seems Chandra had been closely observing the working of missionary William Carey's 12 horsepower steam engine, which was brought from England to run a paper mill. Manufactured by Thwaites, Hick and Rothwell, who were engineers from Bolton, Lancashire, Carey's machine is known to be the first steam engine imported to India. And Goluk Chandra, after studying it, built his own!

It is important to remember that we are talking about the 1820s—a time with no communication, and only primitive transportation methods. In such a scenario, local inventions usually remained local and were quickly forgotten.

THE PORTO NOVO STORY

Let's go back to Porto Novo. It has been documented that in the 1830s, rotary steam engines, patented by William Avery of Syracuse, New York, were used for construction of canals. Avery's steam engine was basically an updated aeolipile, a Heron's engine—a form of steam engine that has been in existence since at least the first century BCE and named after Hero of Alexandria, who came up with the first description of the engine. [7]

These rotary steam engines—widely mentioned as the 'new mechanical agent' or 'machine'—had made their way to the Madras Corps of Engineers and were being experimented with for various uses, including working on railroads as locomotives by General Sir Arthur Thomas Cotton [8] of the Madras Engineers KCSI [9] .

While the stationary steam engines were used to power a lathe, the conclusions deduced from the experiments on these 'mechanical agents' eventually led to the 'construction' of what is documented as the first running steam locomotive in India—at Red Hills near Madras in 1838.